

3D Coordinates Extension — English Worksheet (Beginner)

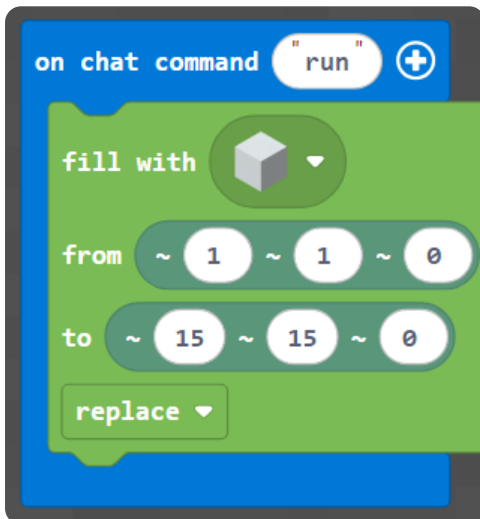
Topic: Pixel-Art Mobs (x, y) · **Course:** 3D Coordinates · **Level:** Extension (Beginner, after Lesson 3) ·

Time: about 30 minutes

A bonus picture challenge: copy a mob onto your wall using **(x, y)** — x is **across**, y is **up**. Read the picture and place each block at its (x, y).

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your home spot:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Stand on the red block every time you run.** If you move, the picture moves too.

1 · Predict 🌟

Read the steps. Circle your answer.

```
place pink block at (4, 13)
place pink block at (5, 13)
place pink block at (6, 13)
```

A line across or up? Circle one: across · up

Which number stays the same? Circle one: x · y

```
place black block at (8, 7)
place black block at (8, 8)
```

Across or up? Circle one: across · up

Which number changes? Circle one: $x \cdot y$

2 · Spot the Bug 🐸

This code is broken. Read what it should do, then fill in the fix.

Bug — The axolotl's two eyes should be at **two different spots**: (6, 10) and (11, 10). Both blocks land in the **same spot**.

```
place black block at (6, 10)
place black block at (6, 10)
```

Hint: the first number (x) of the second eye is wrong. One eye is further **across** than the other, so its x must be **bigger**.

Fill the blank to fix the second line:

```
place black block at (6, 10)
place black block at (__ , 10)
```

Which number fills the blank? Circle one: 6 · 11

Why was the code wrong? Circle one: both eyes had the same x · the color was wrong

3 · Show Your Work 📷 🎥

Stand on your **home spot** (the red block).

Warm up first. Copy this small axolotl frill, reading each pink square's (x, y).

🌸 Axolotl frill

	1	2	3	4	5
5					
4					
3					
2					
1					

Try the big one. Copy the big 🦎 axolotl below, reading each colored square's (x, y).

🦎 Axolotl (pink · #f2a3c0, frills · #ff5fa2, eyes · black)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15															
14															
13															
12															
11															
10															
9															
8															
7															
6															
5															
4															
3															
2															
1															

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.